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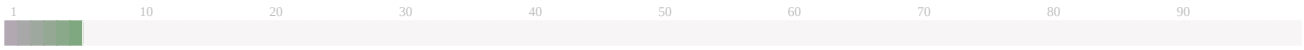
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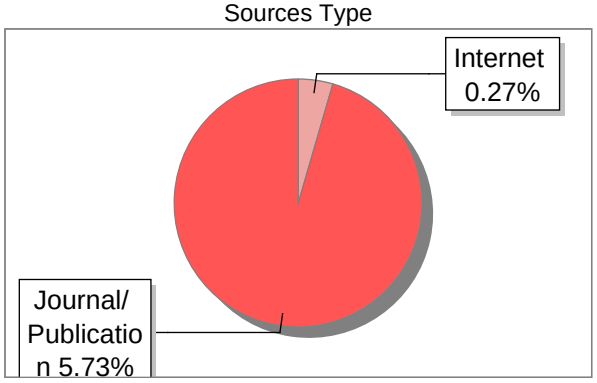
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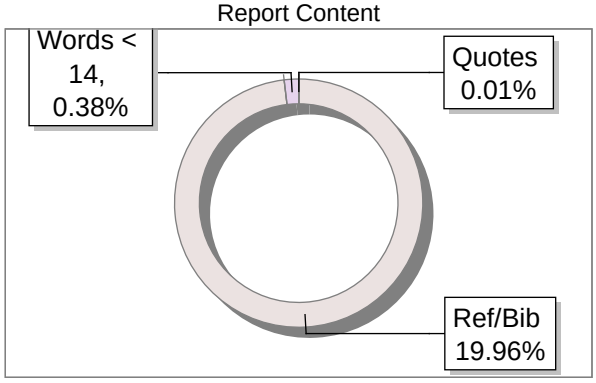
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Time Crystals: From a No-Go Theorem to Driven Phases of Matter

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Spontaneous symmetry breaking is fundamental to the emergence of spatial order in condensed matter and occurs when a ground or thermal state possesses less symmetry than its governing Hamiltonian. Wilczek proposed an analogous breaking of time translation symmetry in which a quantum system in its lowest energy state would exhibit persistent periodic motion. Although this equilibrium proposal was subsequently challenged, the concept of time crystalline order was reformulated in nonequilibrium systems. In periodically driven systems, continuous time translation symmetry is reduced to a discrete symmetry, whose spontaneous breaking can give rise to a *discrete time crystal* (DTC) characterized by robust oscillations with a period equal to an integer multiple of the driving period. Because generic driven systems absorb energy and eventually heat toward infinite temperature, the emergence of such order requires mechanisms, including many-body localization, prethermalization, or integrability that suppress heating. We review the Floquet algebraic origin of long lived oscillations and examine their dependence on driving protocols, initial states, and interactions, while highlighting theoretical and experimental advances and the roles of symmetry, dimensionality, and dissipation in stabilizing DTCs.

Keywords: time crystals; Floquet systems; integrability; dynamical symmetries

I. INTRODUCTION

Symmetry is a central organizing principle of nature. In condensed matter, the spontaneous breaking of symmetry underlies the emergence of order. A ferromagnet breaks spin rotation symmetry, while a crystal breaks continuous spatial translations down to a discrete lattice subgroup. In both cases, the ground state or thermal state exhibits less symmetry than the Hamiltonian from which it arises. The search for new phases of matter, and for the symmetries they break, is central to the development of condensed matter physics and to technological progress. Since relativity places space and time on an equal footing, it is natural to ask whether time-translation symmetry can be broken in the same way.

In 2012, Wilczek and Shapere posed this question for quantum (Wilczek, 2012) and classical (Shapere & Wilczek, 2012) systems, respectively, and introduced the term *time crystal*. Their proposal envisioned a quantum system in its lowest energy state undergoing perpetual periodic motion, such as a rotating density modulation of interacting charged bosons on a ring threaded by magnetic flux. Although this motion would not permit energy extraction, it appeared inconsistent with the conventional understanding that equilibrium states are stationary. The proposal was soon challenged: Bruno (2013) showed that the persistent motion envisaged in Wilczek's ring model cannot occur in a ground state, while Watanabe and Oshikawa (2015) established a general no-go theorem. Since every eigenstate and Gibbs state is stationary at the level of one-point observables, time-translation symmetry breaking (TTSB), like conventional spontaneous symmetry breaking, must be identified through correlation functions. Watanabe and Oshikawa therefore characterized TTSB using the correlator of an extensive observable $\hat{O} = \int d^d x\, \hat{O}(\mathbf{x})$,

$$f(t) = \lim_{V\to\infty} \frac{1}{V^2} \langle e^{i\hat{H}t} \hat{O}\, e^{-i\hat{H}t} \hat{O} \rangle, \tag{1}$$

and showed that $f(t)$ is time independent in the ground state and in Gibbs states of local Hamiltonians. The Gibbs state result was later established rigorously (Watanabe et al., 2020). Time crystals are therefore forbidden in equilibrium.

The idea of TTSB was revived by considering quantum systems driven far from equilibrium. In a periodically driven many-body quantum system, the Hamiltonian satisfies $\hat{H}(t+T) = \hat{H}(t)$, and only the *discrete* time-translation symmetry $t \to t+T$ remains. If the state and the corresponding observables respond with a period mT, where $m > 1$ is an integer, this discrete symmetry is spontaneously broken. This is the essence of a *discrete time crystal* (DTC) (Else et al., 2016; Khemani et al., 2016; Sacha, 2015; von Keyserlingk et al., 2016; Yao et al., 2017). The response must also be robust against perturbations, accompanied by spatial long-range order, and persist in the thermodynamic limit. The main obstacle is that generic interacting driven systems absorb energy and heat to a featureless infinite-temperature state. The first proposals avoided this through disorder-induced many-body localization, while prethermalization at high driving frequency and long-range interactions were later identified as disorder-free routes (Else et al., 2017; Pizzi et al., 2021; Russomanno et al., 2017). Soon after these

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proposals, signatures of period doubling were reported in trapped ions (Zhang et al., 2017) and in dense ensembles of nitrogen-vacancy (NV) centers (Choi et al., 2017). These were followed by observations in NMR systems (Pal et al., 2018; Rovny et al., 2018), nuclear-spin registers (Randall et al., 2021), prethermal trapped-ion chains (Kyprianidis et al., 2021), and superconducting quantum processors (Mi et al., 2022). Comprehensive reviews of the field are available (Else et al., 2020; Khemani et al., 2019; Sacha & Zakrzewski, 2018; Zaletel et al., 2023), together with a monograph (Sacha, 2020).

In this short review, we focus on the Floquet algebraic origin of long-lived oscillations and their dependence on driving protocols, initial states, and interactions. We highlight theoretical and experimental advances and the roles of symmetry, dimensionality, and dissipation in stabilizing DTCs. We restrict our attention to disorder-free spin-1/2 systems, both interacting and integrable, where many body localization is unavailable and robustness must come from other mechanisms. We also discuss the role of integrability, which suppresses heating and can sustain persistent oscillations, and ask when these oscillations survive integrability breaking perturbations and so constitute genuine time crystalline order. Section II introduces the Floquet formalism, the criteria for time crystalline order, and the main stabilization mechanisms. Section III examines integrability and the dynamical symmetries that underlie persistent oscillations. Section IV reviews experimental realizations, and Sec. V discusses open problems.

II. DISCRETE TIME CRYSTALS IN FLOQUET SYSTEMS

A. Floquet formalism and criteria for time-crystalline order

With the no-go theorem, the search for time crystals shifted to nonequilibrium systems. The first proposals were for periodically driven (Floquet) systems, which retain only a discrete time translation symmetry. To break this symmetry, the system must respond with a period mT that is an integer multiple $m > 1$ of the drive period T. To understand how such a response can arise, we first review the Floquet formalism.

For a Hamiltonian with period $T = 2\pi/\omega_D$, $\hat{H}(t+T) = \hat{H}(t)$, the wavefunction $|\psi(t)\rangle$ obeys the Schrödinger equation $i\partial_t |\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle$ (we set $\hbar = 1$). The stroboscopic evolution, which determines the observables at times $t = nT$, is generated by the Floquet operator

$$\hat{U}_F \equiv \hat{U}(T,0) = \mathcal{T}\exp\left[-i\int_0^T \hat{H}(t)\,dt\right] = e^{-i\hat{H}_F T}, \qquad |\psi(nT)\rangle = \hat{U}_F^n |\psi(0)\rangle\,, \tag{2}$$

where $\mathcal{T}$ denotes time ordering and $\hat{H}_F$ is the Floquet Hamiltonian. By Floquet's theorem, the solutions of the Schrödinger equation can be written as $|\psi_\alpha(t)\rangle = e^{-i\mu_\alpha t}|\phi_\alpha(t)\rangle$, where $|\phi_\alpha(t+T)\rangle = |\phi_\alpha(t)\rangle$ and the quasienergies μ_α are defined modulo ω_D. The Floquet eigenstates $|\phi_\alpha(0)\rangle$ are invariant under $\hat{U}_F$ up to a phase. They therefore maintain the discrete time translation symmetry $t \to t+T$, and play the role that energy eigenstates and Gibbs states play in equilibrium. Any breaking of this symmetry must therefore be spontaneous.

Discrete time translation symmetry is spontaneously broken if, for generic short range correlated initial states $|\psi(0)\rangle$ (for instance product states), the stroboscopic expectation value $f(nT) = \langle\psi(nT)|\hat{\mathcal{O}}|\psi(nT)\rangle$ of some local observable $\hat{\mathcal{O}}$ is not T-periodic but repeats only after $m > 1$ periods,

$$f\big((n+1)T\big) \;/\; \qquad f\big((n+m)T\big) = f(nT). \tag{3}$$

This manifests as a subharmonic response, which can be readily observed as a peak at frequency ω_D/m in the Fourier spectrum of $f(nT)$. Equivalently, the Floquet eigenstates are long range correlated cat states. Their quasienergies are organized in multiplets split by ω_D/m, which is the Floquet analogue of the correlator criterion in Eq. (1) (Else et al., 2016; Khemani et al., 2017; von Keyserlingk et al., 2016).

A subharmonic response alone, however, does not guarantee a discrete time crystal. The response must emerge from the collective dynamics of the interacting driven system. Non-interacting spins subjected to perfect π pulses flip every period and trivially show period doubling. A spin rotation error ε_r, however, shifts the response frequency from π/T to $\pi(1-\varepsilon_r)/T$, so the period is not locked. To label a system a discrete time crystal, it must therefore satisfy the following criteria (Else et al., 2016; Khemani et al., 2016; Yao et al., 2017):

a) The system should exhibit spontaneous breaking of discrete time translation symmetry: for generic short-range correlated initial states, the stroboscopic expectation value of a local observable should satisfy Eq. (3), even though the Hamiltonian is periodic, $\hat{H}(t) = \hat{H}(t+T)$.

b) The response should be rigid: the subharmonic oscillations should remain locked at period mT over a finite range of parameters, including imperfect pulses, without fine tuning the system parameters.

c) The oscillating order parameter should be long-range correlated in space, and the oscillations should persist indefinitely in the thermodynamic limit. The limits must be taken in the order $\lim_{nT\to\infty}\lim_{N\to\infty}$, since in any finite system tunnelling between the cat state partners eventually destroys the oscillation.

Therefore, the dynamics of a periodically driven interacting system must support a robust subharmonic response that persists in the thermodynamic limit. In generic driven systems, however, energy absorption from the drive leads to heating toward infinite temperature.

Because a periodic drive conserves no energy, the Floquet eigenstates of a generic interacting system satisfy a Floquet version of the eigenstate thermalization hypothesis (ETH) (D'Alessio et al., 2016; Deutsch, 1991; Srednicki, 1994). Local observables then relax to their infinite-temperature values, which destroys the subharmonic response (D'Alessio & Rigol, 2014; Lazarides et al., 2014a). A central challenge is therefore to identify mechanisms that suppress heating while stabilizing the subharmonic dynamics. One widely studied approach is to localize the many-body system throughout the drive cycle. Other approaches slow down or prevent heating without localization.

Several theoretically and experimentally established mechanisms can provide such localization or otherwise suppress heating. The most widely used approach is to introduce strong disorder, which can produce many-body localization (MBL) (Abanin et al., 2019; Basko et al., 2006; Pal & Huse, 2010). An MBL system possesses an extensive set of quasi-local integrals of motion (Huse et al., 2014; Serbyn et al., 2013), and MBL has been proven in one-dimensional spin chains under an assumption on level statistics (Imbrie, 2016). MBL persists under periodic driving (Lazarides et al., 2015; Ponte et al., 2015). Its quasi-local integrals of motion inhibit energy absorption from the drive and thus prevent heating to infinite temperature. MBL is stable against weak perturbations within the isolated system. It is, however, destroyed by coupling to an environment, and its stability in higher dimensions, with long-range interactions, and asymptotically even in one dimension remains under active debate (De Roeck & Huveneers, 2017; Morningstar et al., 2022; Šuntajs et al., 2020).

A second approach is to use a clean, disorder-free system driven at high frequency. When the drive frequency is large compared with the local energy scale $J_{\rm loc}$, heating is exponentially slow, with a rate bounded by $e^{-c\,\omega_D/J_{\rm loc}}$ (Abanin et al., 2017; Kuwahara et al., 2016; Mori et al., 2016). The system then enters a prethermal regime in which its dynamics are governed, for long but finite times, by an effective time-independent Hamiltonian before eventual heating occurs. By the Watanabe-Oshikawa theorem, such a static effective Hamiltonian alone cannot host a time crystal. For a drive containing a near-π global spin flip $\hat{P}$, however, the Floquet operator takes the form $\hat{U}_F \approx \hat{P}\, e^{-i\hat{D}T}$, where the effective Hamiltonian $\hat{D}$ has an emergent $\mathbb{Z}_2$ symmetry, $[\hat{D},\hat{P}]=0$ (Else et al., 2017). If the prethermal state of $\hat{D}$ spontaneously breaks this symmetry, the system shows period doubling. This requires a finite-temperature ordered phase, and the oscillation persists up to the exponentially long heating time rather than strictly forever.

Additionally, precise engineering of the drive protocol can strongly suppress thermalization. At special drive parameters, two-tone drive protocols manifest exactly flat Floquet bands and thereby suppress heating (Banerjee et al., 2025). More generally, the leading-order Floquet Hamiltonian can acquire emergent conservation laws at special drive amplitudes and frequencies, which constrains the dynamics and produces long-lived nonthermal behavior.

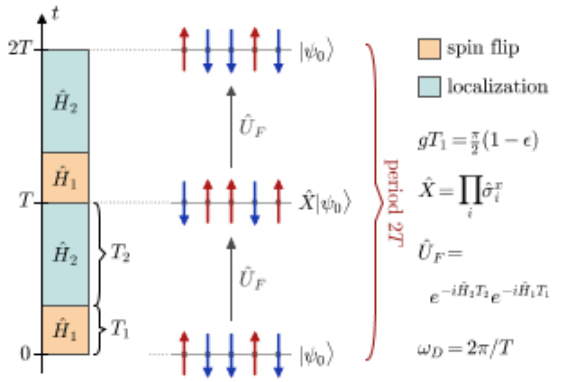


FIG. 1. Schematic of the discrete time crystal drive protocol. The system evolves under two Hamiltonians, $\hat{H}_1$ and $\hat{H}_2$, for durations T_1 and $T_2 = T - T_1$, respectively. The first step, $\hat{H}_1 = \frac{\pi}{2T_1}(1-\varepsilon_r)\sum_j \hat{\sigma}_j^x$, implements an approximately global spin flip with spin-rotation error ε_r, while the second is generated by an interacting local or quasi-local Hamiltonian. Over a finite range of ε_r, the resulting stroboscopic magnetization exhibits a subharmonic response locked at period $2T$.

A further approach is to consider an integrable system whose dynamics are constrained by an extensive set of conserved quantities. Such a system relaxes to a generalized Gibbs ensemble rather than to infinite temperature. Under appropriate conditions, integrability can then support persistent oscillations. These arise when the dynamics admits a dynamical symmetry, an operator $\hat{A}$ satisfying $\hat{U}_F^\dagger \hat{A}\, \hat{U}_F = e^{i\theta}\hat{A}$, for which $\theta = \pi$ gives period doubling. In isolated Hamiltonian systems, the analogous condition $[\hat{H},\hat{A}] = \omega \hat{A}$ sustains undamped oscillations (Buča et al., 2019; Medenjak et al., 2020). For the kicked Ising chain of Sec. II B at perfect π pulses, $\hat{A} = \hat{\sigma}_j^z$ satisfies this condition with $\theta = \pi$ on every site. Each of the mechanisms above can then be viewed as a way of ensuring that a dressed version of $\hat{\sigma}_j^z$ survives pulse errors. We return to dynamical symmetries in Sec. III A.

Section II B introduces the generic drive protocol for a discrete time crystal, shown schematically in Fig. 1, and reviews established results on the mechanisms that produce robust discrete time crystalline order.

B. Drive protocols and stabilization mechanisms

The canonical protocol for realizing a discrete time crystal is based on a binary Floquet drive, shown schematically in Fig. 1. During each period, the system evolves under two Hamiltonians, $\hat{H}_1$ and $\hat{H}_2$, for durations T_1 and $T_2 = T - T_1$, respectively. Defining

$$\hat{U}_1 = e^{-i\hat{H}_1 T_1}, \qquad \hat{U}_2 = e^{-i\hat{H}_2 T_2}, \tag{4}$$

the one-period Floquet operator is

$$\hat{U}_F = \hat{U}_2\hat{U}_1 = e^{-i\hat{H}_2 T_2}\, e^{-i\hat{H}_1 T_1}. \tag{5}$$

The first step is chosen to implement an approximately global spin flip, while the second is generated by an interacting local or quasi-local Hamiltonian. For a spin-1/2 chain, a representative choice is (Yao et al., 2017)

$$\begin{aligned} \hat{H}_1 &= g\sum_i \hat{\sigma}_i^x, \\ \hat{H}_2 &= \sum_i \left(J_i\hat{\sigma}_i^z\hat{\sigma}_{i+1}^z + h_i\hat{\sigma}_i^z\right), \end{aligned} \tag{6}$$

where J_i and h_i denote random couplings and longitudinal fields. In the limit $T_1 \to 0$ with gT_1 fixed, the first step becomes an instantaneous kick, and the protocol is known as the kicked Ising chain. Experimentally, the global spin flip is implemented by a nominal π pulse, which is generally imperfect. We parametrize the pulse imperfection by

$$g = \frac{\pi}{2T_1}\left(1-\varepsilon_r\right), \tag{7}$$

so that $\hat{U}_1$ rotates every spin about the x axis by the angle $\pi(1-\varepsilon_r)$. The spin rotation error ε_r vanishes for ideal pulses.

For an ideal pulse, $\varepsilon_r = 0$, the first step reduces to $\hat{U}_1 = (-i)^N\hat{P}$ with $\hat{P} = \prod_i \hat{\sigma}_i^x$, and flips every spin,

$$\hat{U}_1^\dagger\hat{\sigma}_i^z\hat{U}_1 = -\hat{\sigma}_i^z, \qquad \hat{U}_1^\dagger\Big(\sum_i \hat{\sigma}_i^z\Big)\hat{U}_1 = -\sum_i \hat{\sigma}_i^z. \tag{8}$$

The stroboscopic magnetization is

$$M(nT) = \frac{1}{N}\sum_i \langle\psi_0|(\hat{U}_F^\dagger)^n\,\hat{\sigma}_i^z\,\hat{U}_F^n|\psi_0\rangle. \tag{9}$$

Since $\hat{H}_2$ commutes with every $\hat{\sigma}_i^z$, at $\varepsilon_r = 0$ we have $\hat{U}_F^\dagger\hat{\sigma}_i^z\hat{U}_F = -\hat{\sigma}_i^z$. This is precisely the dynamical symmetry with $\theta = \pi$ introduced in Sec. II A and discussed further in Sec. III A. The magnetization therefore alternates between successive periods,

$$M(nT) = (-1)^n M(0), \qquad M\big((n+2)T\big) = M(nT). \tag{10}$$

Thus, the observable responds with period $2T$, rather than with the drive period T, confirming the subharmonic response associated with discrete time translation symmetry breaking. At this fine tuned point, however, the period doubling is trivial, since it holds equally in the absence of interactions. In a genuine discrete time crystalline phase, the response must remain stable against perturbations and persist indefinitely in the thermodynamic limit.

The defining feature of the discrete time crystalline phase is that the subharmonic response remains stable over a finite interval of ε_r, rather than occurring only at the fine-tuned value $\varepsilon_r = 0$. This robustness arises from

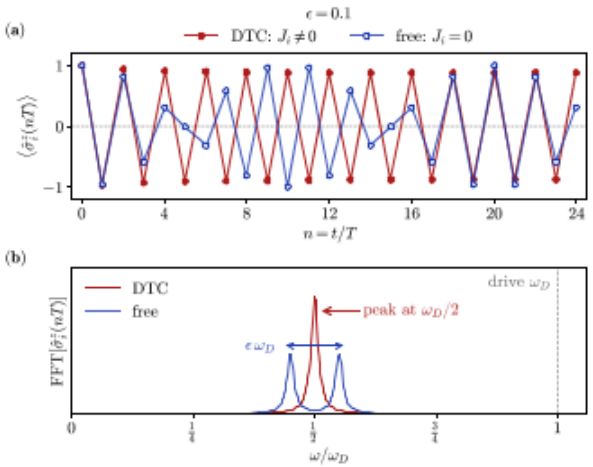


FIG. 2. (a) Stroboscopic magnetization $M(nT)$, Eq. (9), at $\varepsilon_r = 0.1$ for interacting ($J_i > 0$) and non-interacting ($J_i = 0$) chains. The interacting chain shows period doubling locked at $2T$, while the non-interacting chain shows beating. (b) Fourier spectrum of $M(nT)$. With interactions the peak is pinned at $\omega = \pi/T$; without interactions it shifts to $\omega = \pi(1-\varepsilon_r)/T$.

the interacting dynamics during the T_2 interval, provided that heating is suppressed and the time crystalline order is preserved. Fig. 2 illustrates this at $\varepsilon_r = 0.1$. In the non-interacting limit, $J_i = 0$ (with $h_i = 0$), each spin simply rotates by $\pi(1-\varepsilon_r)$ per period, and the Fourier peak of the magnetization shifts from $\omega = \pi/T$ to $\omega = \pi(1-\varepsilon_r)/T$. In the presence of interactions, $J_i > 0$, the peak remains sharply locked at $\omega = \pi/T$. The interaction term thus counteracts the phase error introduced by the finite spin-rotation error ε_r. It is therefore crucial for both the emergence and the stabilization of the subharmonic response (Else et al., 2016; Khemani et al., 2016; Yao et al., 2017).

The origin of this locking can be understood from the structure of $\hat{U}_F$. At $\varepsilon_r = 0$, its eigenstates are cat states $|\{s\}\rangle \pm e^{i\phi_s}|\{-s\}\rangle$ built from $\hat{\sigma}^z$ product states, with quasienergies differing by exactly π/T. A small rotation error flips individual spins and therefore creates domain walls, which cost an energy of order J_i. When the couplings and fields are random, these processes are generically off-resonant, so ε_r acts only perturbatively. It dresses $\hat{\sigma}_i^z$ into quasi-local operators $\hat{\tau}_i^z$ that still satisfy $\hat{U}_F^\dagger\hat{\tau}_i^z\hat{U}_F \simeq -\hat{\tau}_i^z$, and the π pairing survives with corrections exponentially small in the system size (Else et al., 2016; Khemani et al., 2016; von Keyserlingk et al., 2016). This is the many-body-localized DTC. Its rigidity has been established theoretically (Else et al., 2016; Khemani et al., 2016; von Keyserlingk et al., 2016; von Keyserlingk & Sondhi, 2016; Yao et al., 2017), and signatures of it have been observed experimentally (Mi et al., 2022; Randall et al., 2021).

However, quenched disorder is not essential for realizing

DTC order, and the asymptotic stability of MBL is itself under debate. We therefore turn to clean periodically driven systems, where dynamical many body localization (DMBL) can suppress heating through the intrinsic dynamics and provide a route toward time crystalline behavior in the absence of disorder. A representative example is a long-range Ising chain subject to a periodically modulated field,

$$\hat{H}_2(t) = \sum_{i<j} J_{ij}\,\hat{\sigma}_i^y\hat{\sigma}_j^y - h\sin(\Omega t)\sum_i \hat{\sigma}_i^z, \qquad J_{ij} = \frac{J_0}{|i-j|^\alpha}, \tag{11}$$

where Ω is the frequency of the field modulation. As demonstrated by Rahaman et al. (2024a,b), this model exhibits robust dynamical localization when the drive parameters satisfy $\mathcal{J}_0\,(4h/\Omega) = 0$, where $\mathcal{J}_0$ denotes the zeroth-order Bessel function of the first kind. The origin of this condition is clearest in the frame rotating with the field. Writing $\hat{\sigma}_i^y\hat{\sigma}_j^y$ in terms of $\hat{\sigma}^\pm$, the flip-flop terms $\hat{\sigma}_i^+\hat{\sigma}_j^- + \text{h.c.}$ conserve $\sum_i \hat{\sigma}_i^z$ and are unaffected by the rotation. The pair creation terms $\hat{\sigma}_i^+\hat{\sigma}_j^+ + \text{h.c.}$ instead acquire, up to a constant phase, the time dependent phase $e^{\pm i(4h/\Omega)\cos\Omega t}$. Averaged over a cycle, they are multiplied by $\mathcal{J}_0(4h/\Omega)$ and therefore vanish at its zeros. To leading order in $1/\Omega$, the effective Hamiltonian then conserves the total magnetization, and the resulting dynamical localization suppresses heating (Rahaman et al., 2024a). This localization mechanism can be combined with the binary Floquet protocol of Eq. (5), with $\hat{H}_2(t)$ acting during the interval T_2 (chosen as an integer multiple of $2\pi/\Omega$), to stabilize a subharmonic response and thereby realize DTC order in a clean system (Rahaman et al., 2024b).

A complementary route to realizing DTC order without disorder is provided by high-frequency driving. When the drive frequency ω_D is much larger than the local energy scale $J_{\rm loc}$, absorbing a single drive quantum requires a high-order many-body process. Consequently, the heating time is exponentially long (Abanin et al., 2017; Kuwahara et al., 2016; Mori et al., 2016),

$$\tau_{\rm heat} \gtrsim \exp\big(c\,\omega_D/J_{\rm loc}\big). \tag{12}$$

For times shorter than $\tau_{\rm heat}$, a suitable rotating frame transformation brings the Floquet operator to the form $\hat{U}_F \approx \hat{P}\,e^{-i\hat{D}T}$. The effective static Hamiltonian $\hat{D}$ has an *emergent* $\mathbb{Z}_2$ symmetry, $[\hat{D},\hat{P}] = 0$. If the corresponding prethermal state spontaneously breaks this symmetry, the discrete time translation symmetry is broken and the laboratory frame observable exhibits a robust period-doubled response (Else et al., 2017). Thus, the existence of a prethermal DTC is constrained by the same equilibrium symmetry breaking considerations discussed above. For short range interactions, finite temperature breaking of a discrete symmetry requires $d \geq 2$. Clean long range interactions can modify this conclusion and support time crystalline order even in one dimension (Pizzi et al., 2021; Russomanno et al., 2017). A prethermal DTC of this kind has been observed in a trapped-ion chain (Kyprianidis et al., 2021).

Another route to realizing DTC order in a clean system is provided by a two-tone drive protocol that dynamically cancels the interaction contribution to the Floquet Hamiltonian, resulting in flat quasienergy bands (Banerjee et al., 2025). In this scheme, the system is subjected to two square wave drives with commensurate frequencies ω_1 and ω_2, satisfying $\omega_1 = n\omega_2$, where $n > 1$ is an odd integer. The corresponding time-dependent Hamiltonian is

$$\hat{H}_2(t) = J\sum_i \hat{\sigma}_i^x\hat{\sigma}_{i+1}^x + h\Big[\operatorname{sgn}\big(\sin\omega_1 t\big) - \operatorname{sgn}\big(\sin\omega_2 t\big)\Big]\sum_i \hat{\sigma}_i^z, \tag{13}$$

where sgn denotes the sign function. For the commensurate choice of drive frequencies specified above, the two-tone protocol cancels the interaction contribution over a driving cycle. The resulting flat quasienergy bands produce dynamical localization and thereby suppress heating. When combined with a binary Floquet protocol that includes a spin flip operation, this localization mechanism stabilizes a subharmonic response. If the cancellation were exact, however, the evolution during T_2 would act trivially and could not correct spin-rotation errors. As in Fig. 2, the response would then not be locked. Small residual interactions restore this rigidity and stabilize the response against spin-rotation errors, which act as external perturbations. Rahaman and Roy (2026) showed that Floquet flat bands of this kind enable discrete time crystalline order in spin chains, with a robust period doubled stroboscopic magnetization. Thus, this protocol provides a route to realizing clean DTC order without relying on quenched disorder.

III. INTEGRABILITY AND THE (USUAL) ABSENCE OF OSCILLATIONS

An integrable many-body system has an extensive set of local or quasi-local conserved charges $\{\hat{Q}_j\}$ with $[\hat{H},\hat{Q}_j] = [\hat{Q}_j,\hat{Q}_l] = 0$ (Essler & Fagotti, 2016; Ilievski et al., 2016). Free-fermion models such as the transverse-field Ising chain (Lieb et al., 1961; Pfeuty, 1970) belong to this class, as do interacting Bethe-ansatz-solvable models such as the XXZ and Hubbard chains. Because these charges constrain the dynamics, integrable systems do not thermalize. After a quench they relax locally to a generalized Gibbs ensemble (GGE) (Ilievski et al., 2015; Rigol et al., 2007; Vidmar & Rigol, 2016),

$$\hat{\rho}_{\rm GGE} = \frac{1}{Z}\exp\Big(-\sum_j \beta_j\hat{Q}_j\Big), \tag{14}$$

where the Lagrange multipliers β_j are fixed by the initial values $\langle \hat{Q}_j \rangle$. The quantum Newton's cradle (Kinoshita et al., 2006) demonstrates this memory of initial conditions. Periodically driven integrable systems similarly relax to a periodic GGE (Lazarides et al., 2014b).

Note that integrable systems *do* relax, just not to a thermal state. Expand the state in energy eigenstates, $|\psi\rangle = \sum_m c_m |m\rangle$. Then

$$\langle \hat{O}(t) \rangle = \sum_m |c_m|^2 O_{mm} + \sum_{m \neq n} c_m^* c_n \, O_{mn} \, e^{i(E_m - E_n)t}. \tag{15}$$

In the thermodynamic limit, the energy differences $E_m - E_n$ form a continuum. When the weights $c_m^* c_n O_{mn}$ are smoothly distributed over it, the second term dephases by the Riemann-Lebesgue lemma, leaving only the diagonal (GGE) part. For a periodically driven system, the same argument applies at stroboscopic times $t = nT$, with energies replaced by quasienergies. Conserved charges with finite overlap with an observable bound the time average of its autocorrelation function from below (Mazur, 1969; Suzuki, 1971), but this bound concerns the stationary part of the response rather than oscillations. Conservation laws therefore determine *where* the system relaxes, but they do not by themselves prevent relaxation. Persistent oscillations require more: the spectrum must contain an extensive set of exactly commensurate energy differences, and these differences must contribute finite weight to local observables. Section III A explains what algebraic structure gives rise to such commensurability, and Sec. III B discusses its realization in free-fermion models.

A. Dynamical symmetries

The algebraic structure that produces exactly commensurate energy differences is a dynamical symmetry (Buča et al., 2019; Medenjak et al., 2020). An operator $\hat{A}$ is a dynamical symmetry of a time-independent Hamiltonian if

$$[\hat{H}, \hat{A}] = \omega \hat{A}, \tag{16}$$

with ω real. It follows that $\hat{A}(t) = e^{i\hat{H}t} \hat{A} e^{-i\hat{H}t} = e^{i\omega t} \hat{A}$, so that $\langle \hat{A}(t) \rangle = e^{i\omega t} \langle \hat{A}(0) \rangle$ for every initial state, and the oscillation never decays. In spectral terms, if $\hat{H}|E\rangle = E|E\rangle$, then $\hat{H}\hat{A}|E\rangle = (E + \omega)\hat{A}|E\rangle$. The operator $\hat{A}$ therefore connects eigenstates whose energies differ by exactly ω. When $\hat{A}$ is a sum of local terms, it supplies precisely the extensive set of commensurate energy differences with finite weight that Eq. (15) requires.

The simplest example is the isotropic Heisenberg chain in a uniform field, $\hat{H} = J \sum_i \hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_{i+1} + h \sum_i \hat{S}_i^z$. Because the exchange term is invariant under global spin rotations, the total raising operator satisfies $[\hat{H}, \hat{S}^+_{\rm tot}] = h \, \hat{S}^+_{\rm tot}$, and the total transverse magnetization precesses at frequency h without decay. This does not contradict the Watanabe-Oshikawa theorem: $\langle \hat{S}^+_{\rm tot} \rangle$ vanishes in every Gibbs state of $\hat{H}$, so the oscillation requires a nonequilibrium initial state. A second example is Yang's η-pairing operator in the Hubbard model on a bipartite lattice, which satisfies Eq. (16) with a frequency set by the on-site interaction and the chemical potential (Yang, 1989). Medenjak et al. (2020) argued that isolated Heisenberg magnets with such structure can be regarded as quantum time crystals. Long-lived phantom-helix states, whose longevity reflects exact properties of the anisotropic Heisenberg model, have been observed with ultracold atoms (Jepsen et al., 2022). In all these cases, however, the frequency is set by Hamiltonian parameters and shifts continuously with them. Such oscillations are therefore not rigid in the sense of criterion (b).

For a periodically driven system, the analogue of Eq. (16) is

$$\hat{U}_F^\dagger \hat{A} \hat{U}_F = e^{i\theta} \hat{A}, \tag{17}$$

which gives $\hat{A}(nT) = e^{in\theta} \hat{A}$. A phase $\theta = 2\pi/m$ produces a response with period mT, and $\theta = \pi$ gives period doubling. In spectral terms, $\hat{A}$ connects Floquet eigenstates whose quasienergies differ by θ/T, which for $\theta = \pi$ is the π pairing of Sec. II B. In the kicked Ising chain at $\varepsilon_r = 0$, every $\hat{\sigma}_i^z$ satisfies Eq. (17) with $\theta = \pi$. Here the phase is not set by a tunable Hamiltonian parameter but by the $\mathbb{Z}_2$ structure of the drive, $\hat{P}^2 = \hat{\mathbf{1}}$. As long as weak perturbations preserve the dressed form $\hat{U}_F \approx \hat{P} \, e^{-i\hat{D}T}$ with $[\hat{D}, \hat{P}] = 0$, the phase $\theta = \pi$ cannot drift continuously. This is the algebraic origin of the rigidity of the period in a discrete time crystal, and it distinguishes Floquet dynamical symmetries from their Hamiltonian counterparts. Floquet dynamical symmetries of this type have been used to construct time crystals in driven Hubbard models (Chinzei & Ikeda, 2020). The free-fermion models of Sec. III B provide a further example, in which the condition is met for selected momenta.

B. Free-fermion models

Time-crystalline responses can be realized in a class of integrable free-fermion models (Chandra & Roy, 2024),

$$\begin{aligned} \hat{H}(t) &= \sum_{\boldsymbol{k}>0} \hat{H}^{(0)}_{\boldsymbol{k}}, \\ \hat{H}^{(0)}_{\boldsymbol{k}} &= \big(a_0 - b_{\boldsymbol{k}}\big)\big(n_{\boldsymbol{k}} + n_{-\boldsymbol{k}}\big) + D_{\boldsymbol{k}}\big(c^\dagger_{\boldsymbol{k}} c^\dagger_{-\boldsymbol{k}} + \text{h.c.}\big), \end{aligned} \tag{18}$$

with $n_{\pm\boldsymbol{k}} = c^\dagger_{\pm\boldsymbol{k}} c_{\pm\boldsymbol{k}}$. This Cooper-pair Hamiltonian is specified by the parameters a_0, $b_{\boldsymbol{k}}$, and $D_{\boldsymbol{k}}$ and by the fermionic ladder operators $c_{\pm\boldsymbol{k}}$ and $c^\dagger_{\pm\boldsymbol{k}}$. It consists of a kinetic term with coupling $a_0 - b_{\boldsymbol{k}}$ and a pairing term with coupling $D_{\boldsymbol{k}}$, where a_0 is time dependent. Consider driving the system with a square-wave signal of period

T and a 50% duty cycle, such that the first half-cycle flips the spins while the second half-cycle does not. The Hamiltonian during the nth cycle is then

$$\hat{H}_{\boldsymbol{k}}(t) = \begin{cases} \hat{H}^{(0)}_{\boldsymbol{k}}, & nT \le t < \left(n+\frac{1}{2}\right)T, \\ a_1\left(c^\dagger_{\boldsymbol{k}} c^\dagger_{-\boldsymbol{k}} + \text{h.c.}\right), & \left(n+\frac{1}{2}\right)T \le t < (n+1)T. \end{cases} \tag{19}$$

For a particular momentum, $\boldsymbol{k} = \boldsymbol{k}_0$, if $a_0 = b_{\boldsymbol{k}_0}$ and $\omega_D \equiv 2\pi/T = 2D_{\boldsymbol{k}_0}$, the wave function repeats after a time $2T$. This subharmonic response is a hallmark of discrete time-crystalline behavior.

For the one-dimensional (1D) transverse-field Ising model (TFIM), the subharmonic conditions reduce to $a_0 = b_{k_0} = \cos k_0$ and $\omega_D = 2\sin k_0$. However, a small deviation of the momentum from the specific value k_0 destroys the time-crystalline response. In the two-dimensional (2D) TFIM, by contrast, the conditions become $a_0 = b_{\boldsymbol{k}} = \cos k_x + \alpha \cos k_y$ and $\omega_D = 2\sin k_x + 2\alpha \sin k_y$. Setting $\alpha = 0$ recovers the 1D case. The additional tunable parameter in two dimensions therefore permits a continuous range of momenta supporting a stable time crystalline response. In the 1D TFIM, no analogous parameter is available, precluding the construction of a stable time crystal by this mechanism.

To obtain stable time-crystalline responses in 1D, one can supplement the TFIM with a next-nearest-neighbor (NNN), or equivalently three-spin, interaction. The model remains integrable, while the subharmonic conditions become $a_0 = b_k = \cos k + \lambda \cos 2k$ and $\omega_D = 2\sin k + 2\lambda \sin 2k$, where λ denotes the three-spin coupling. This additional interaction provides the tunability needed to construct a 1D time crystal and overcome the limitations of the 1D TFIM.

IV. EXPERIMENTAL REALIZATIONS

The first observations of discrete time crystalline order used two complementary platforms. Zhang et al. (2017) drove a chain of trapped ions with long range Ising interactions and engineered disorder. They observed a period doubled magnetization that remained locked at $2T$ over a finite range of pulse imperfections and melted when the imperfections became large. Because the chain was finite and the interactions long ranged, it is difficult to establish that the stabilization in this experiment is genuinely due to many body localization. Choi et al. (2017) used a dense ensemble of nitrogen-vacancy (NV) centers in diamond with disordered dipolar interactions. In this system the order is not stabilized by localization; instead, slow, critical thermalization of the dipolar ensemble extends its lifetime. Both experiments demonstrated a rigid subharmonic response, but neither could test persistence to infinite times.

Experiments in clean systems followed, which is the setting of primary interest in this chapter. NMR experiments on star shaped molecular clusters (Pal et al., 2018) and on a spatially ordered dipolar crystal (Rovny et al., 2018) showed signatures of time crystalline order without engineered disorder. The mechanism that stabilizes the response in the dipolar crystal is still debated, with prethermalization and long range interactions as the main candidates. A prethermal discrete time crystal was subsequently demonstrated with trapped ions (Kyprianidis et al., 2021). Its lifetime grows with the drive frequency, as expected from the slow heating of the prethermal regime discussed in Sec. II B. In Rydberg atom arrays (Bluvstein et al., 2021), periodic driving stabilizes period doubled revivals of quantum many-body scarred states (Serbyn et al., 2021; Turner et al., 2018). This experiment shows that subharmonic dynamics can arise from restricted, rather than complete, algebraic structure in the spectrum.

Later experiments addressed eigenstate order directly. Randall et al. (2021) used a programmable register of ^{13}C nuclear spins surrounding an NV center to probe a many body localized discrete time crystal. Mi et al. (2022) used a superconducting quantum processor to show that time-crystalline order persists across the spectrum, including for random initial states. Their echo protocol was designed to separate the intrinsic decay of the order from decoherence. A realization on a large cloud based quantum computer followed (Frey & Rachel, 2022), showing that such Floquet circuits can now be implemented on general purpose quantum hardware.

Time crystalline behaviour has also been observed beyond closed spin systems. Magnon condensates in superfluid ^{3}He exhibit a time quasicrystal and its transition to a time crystal (Autti et al., 2018). In atom cavity systems, dissipation stabilizes a period doubled response (Keßler et al., 2021), and continuous pumping produces self sustained oscillations that break continuous time translation symmetry (Kongkhambut et al., 2022). These open system realizations show that dissipation can stabilize, rather than only destroy, time crystalline order.

Several of these experiments connect directly to the themes of this chapter. Floquet circuits on superconducting processors closely match the kicked Ising chain of Eqs. (5) and (6), with integrability breaking terms that can be tuned. When the longitudinal fields vanish, the circuit reduces to a free fermion model, so the free fermion physics discussed in Sec. III B can be probed directly. The phantom helix experiments in ultracold atoms (Jepsen et al., 2022) probe the dynamical symmetry physics of Sec. III A in an isolated Heisenberg magnet. In all platforms, however, the observation time is limited by decoherence, and hybrid quantum-classical feedback has been proposed as a way to prolong discrete time crystalline order on such hardware (Camacho & Fauseweh, 2024). Separating intrinsic persistence, as required by criterion (c), from experimental limitations therefore remains a central challenge.

V. OPEN PROBLEMS AND OUTLOOK

Integrable models give an exact, analytically controlled view of what a time crystal requires. They show that persistent oscillations need commensurate spectral structure, expressed through dynamical symmetries and the π pairing of Floquet eigenstates (Secs. II B and III A). In the one-dimensional TFIM this structure is confined to isolated momenta (Sec. III B). Interactions are needed to lock the response and to stabilize it against the domain walls created by pulse errors (Sec. II B). Integrable models thus explain why the structure required for a time crystal is generically fragile. Several directions remain open.

The first concerns clean interacting systems. It is not known whether there exists a rigorous, non perturbative example of a bulk discrete time crystal in a clean, short range, one dimensional system that persists indefinitely. Prethermal mechanisms cannot provide such an example, since a discrete symmetry cannot be spontaneously broken at finite temperature in one dimension with short range interactions. Edge modes offer an alternative at the boundary. Floquet Majorana π modes produce persistent period doubling at the ends of driven free fermion chains (Thakurathi et al., 2013), and strong and almost strong zero modes give long lived edge spins in interacting chains (Fendley, 2016; Kemp et al., 2017). Extending such edge modes to interacting Floquet chains provides one route toward long lived boundary oscillations. In the bulk, clean long range interactions in one dimension (Pizzi et al., 2021; Russomanno et al., 2017), or short range interactions in higher dimensions (Else et al., 2017), allow time crystalline order without disorder.

A second direction concerns the robustness of dynamical symmetries. The dynamical symmetries of Heisenberg and Hubbard models discussed in Sec. III A produce persistent oscillations, but their frequencies are set by Hamiltonian parameters and shift continuously with them. They are therefore not rigid in the sense of criterion (b). An important question is whether such symmetries can be made robust, for instance through drive-induced locking, so that their frequencies become rigid. The contrast with the $\mathbb{Z}_2$ structure of the kicked Ising chain suggests that a discrete symmetry of the drive is the essential ingredient.

A third direction lies between the integrable and chaotic limits. Quantum many body scars (Serbyn et al., 2021; Turner et al., 2018) and Hilbert space fragmentation (Sala et al., 2020) generate long lived oscillations from *restricted* algebraic structure. In scarred systems, approximately equally spaced towers of eigenstates play the role of a dynamical symmetry acting within a subspace, and periodic driving can stabilize the resulting revivals (Bluvstein et al., 2021). This similarity suggests that a single algebraic framework may describe non stationary many body dynamics.

Open systems raise a further set of questions. Dissipation can stabilize time crystalline order, as in boundary time crystals (Iemini et al., 2018), in dissipative discrete time crystals without manifest symmetries or disorder (Gambetta et al., 2019), and in driven atom cavity systems (Keßler et al., 2021; Kongkhambut et al., 2022). Algebraic conditions for persistent oscillations and limit cycles under dissipation have been formulated (Buča et al., 2022, 2019). How these open system phases relate to the closed system, eigenstate based definition of a discrete time crystal, and to classical discrete time crystals (Zaletel et al., 2023), remains to be clarified.

Responses beyond period doubling also deserve attention. Higher-order and fractional subharmonic responses (Pizzi et al., 2021), as well as time quasicrystals (Autti et al., 2018), remain largely unexplored in integrable settings, where exact solutions could clarify their spectral origin. Equation (17) with $\theta = 2\pi/m$ provides a natural starting point for such an analysis.

Finally, the asymptotic status of many body localization remains unsettled. Although MBL has been proven in one dimension under an assumption on level statistics (Imbrie, 2016), numerical studies questioning its stability (Suntajs et al., 2020) and the identification of avalanche instabilities (De Roeck & Huveneers, 2017; Morningstar et al., 2022) have reopened the question. Its resolution will determine whether disordered discrete time crystals are true phases of matter or merely extremely long lived crossovers. This question also motivates the disorder free mechanisms emphasized in this review.

Time crystals extend the concept of spontaneous symmetry breaking to systems far from equilibrium. Their defining feature is not oscillation itself but its rigidity, which in Floquet systems originates from a discrete symmetry of the drive that pins the response at a fixed fraction of the drive frequency. Integrable models show most clearly what this requires, namely commensurate spectral structure protected against perturbations, and why generic systems struggle to provide it. Whether clean interacting systems can provide it indefinitely, rather than for exponentially long but finite times, remains the central open question of the field.

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